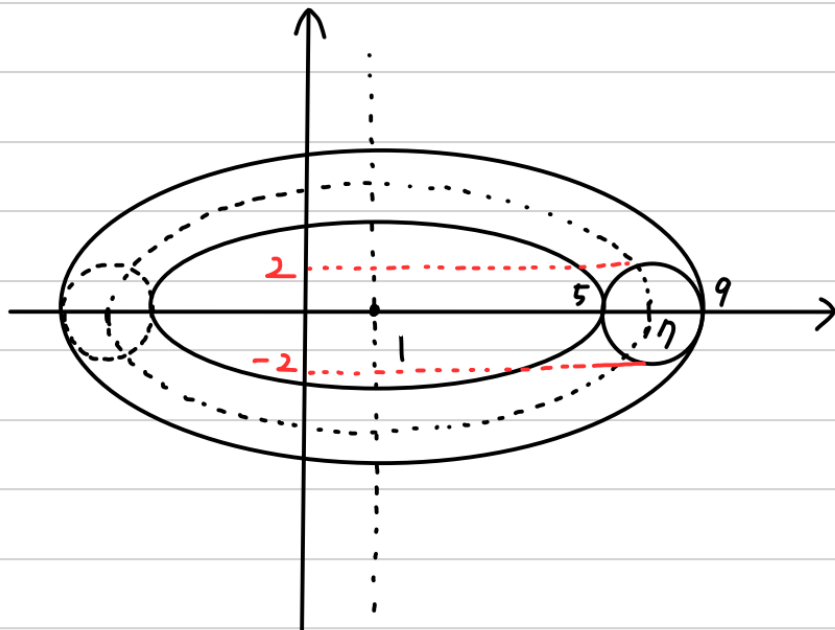


#1

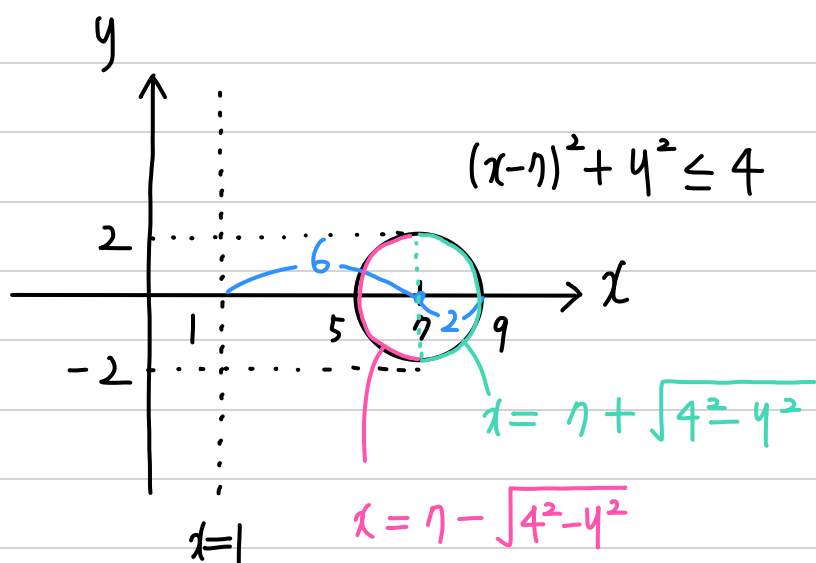
501) ① 압체의 부피 그리기



(p.2에 이어서 서술)

② 입체의 부피 구하기

[방법1]



$$\begin{aligned}
 V &= \pi \int_{-2}^2 (7 + \sqrt{4-y^2} - 1)^2 dy \\
 &\quad - \pi \int_{-2}^2 (7 - \sqrt{4-y^2} - 1)^2 dy \\
 &= \pi \int_{-2}^2 (6 + \sqrt{4-y^2})^2 dy \\
 &\quad - \pi \int_{-2}^2 (6 - \sqrt{4-y^2})^2 dy
 \end{aligned}$$

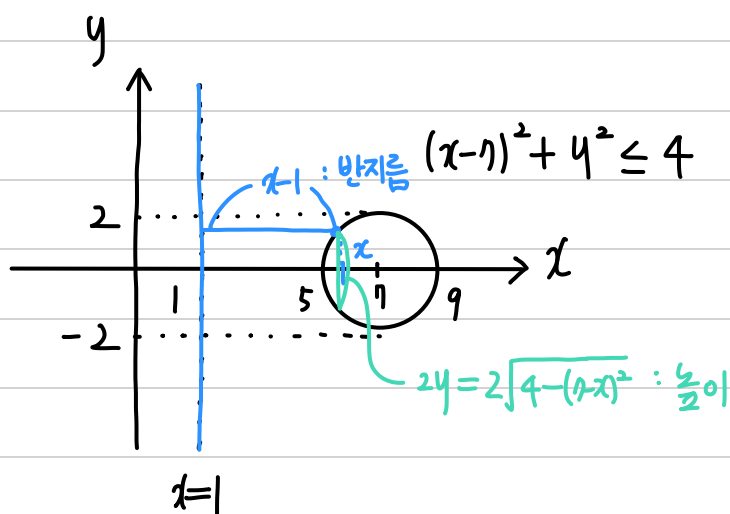
$$= \pi \int_{-2}^2 12 \cdot 2 \sqrt{4-y^2} dy$$

$$= 24\pi \int_{-2}^2 \sqrt{4-y^2} dy$$

$$= 24\pi \cdot (2\pi) = 48\pi^2$$

$$(\because \int_{-2}^2 \sqrt{4-y^2} dy = (\text{반지름} = 2 \text{인 반원의 넓이}) = 2\pi)$$

[방법2] 원통껍질법



$$V = \int_5^9 2\pi(x-1) \cdot 2\sqrt{4-(x-7)^2} dx \text{ 이다.}$$

$$\begin{aligned}
 7-x=t \text{ 치환하면 } &\begin{cases} x=7-t, dx=-dt \\ x=9 \text{ 일때 } t=-2 \\ x=5 \text{ 일때 } t=2 \end{cases} \text{ 이므로}
 \end{aligned}$$

$$\begin{aligned}
 V &= \int_2^{-2} 2\pi(6-t) \cdot 2\sqrt{4-t^2} (-dt) \\
 &= 4\pi \left\{ 6 \int_{-2}^2 \sqrt{4-t^2} dt - \int_{-2}^2 t\sqrt{4-t^2} dt \right\}
 \end{aligned}$$

$$(\because \int_{-2}^2 \sqrt{4-y^2} dy = (\text{반지름} = 2 \text{인 반원의 넓이}) = 2\pi) 2\pi \quad \quad \quad 0 \quad (\because \text{가함수 적분})$$

$$= 48\pi^2$$

#2.

sol) $\sqrt{x} = u$ 로 치환하자. 그러면 $x = u^2$ 이다.

양변을 x 에 대하여 미분하면

$$1 = 2u \frac{du}{dx}, \quad dx = 2u du$$

이다.

$$\int \ln \sqrt{x} dx = \int \ln u \cdot 2u du$$

$$= u^2 \ln u - \int u^2 \cdot \frac{1}{u} du$$

$$= u^2 \ln u - \int u du$$

$$= u^2 \ln u - \frac{1}{2} u^2 + C$$

$$= x \ln \sqrt{x} - \frac{1}{2} x + C$$

#3.

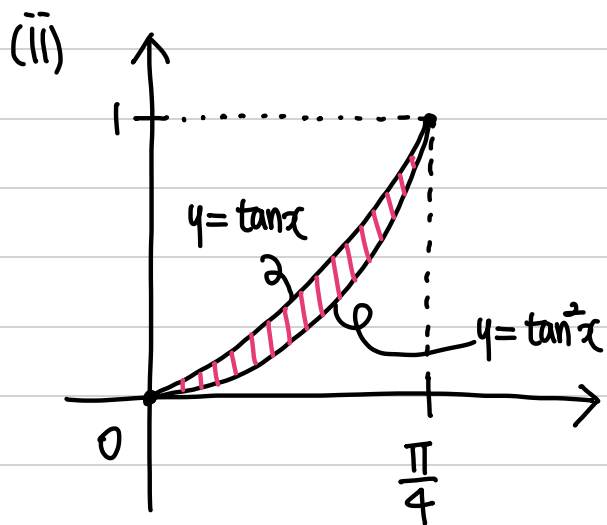
sol) (i) 구간 $[0, \frac{\pi}{4}]$ 에서 두 함수의 교점을 구하자.

$$\tan^2 x = \tan x \Leftrightarrow \tan x (\tan x - 1) = 0$$

$$\Leftrightarrow \tan x = 0 \text{ or } 1$$

$$\therefore x = 0, \frac{\pi}{4}$$

(p.3에 이어서 서술)



즉, 구간 $[0, \frac{\pi}{4}]$ 에서 두 함수로 둘러싸인 넓이는 아래와 같다.

$$\int_0^{\frac{\pi}{4}} |\tan^2 x - \tan x| dx$$

$$= \int_0^{\frac{\pi}{4}} (\tan x - \tan^2 x) dx$$

$$= \int_0^{\frac{\pi}{4}} \tan x dx - \int_0^{\frac{\pi}{4}} \tan^2 x dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} dx - \int_0^{\frac{\pi}{4}} (\sec^2 x - 1) dx$$

$$= \left[-\ln |\cos x| \right]_0^{\frac{\pi}{4}} - \left[\tan x - x \right]_0^{\frac{\pi}{4}}$$

$$= -\ln \frac{\sqrt{2}}{2} - \left(1 - \frac{\pi}{4} \right)$$

$$= \boxed{\frac{1}{2} \ln 2 - 1 + \frac{\pi}{4}} \quad (\because -\ln \frac{\sqrt{2}}{2} = -\ln \frac{1}{\sqrt{2}} = \frac{1}{2} \ln 2)$$